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Research paper

Observer-based fixed-time tracking control for space robots in task space

Rongyu Jin [a], Paolo Rocco [b], Yunhai Geng [a,*]

[a] Research Center of Satellite Technology, Harbin Institute of Technology, Harbin, 150001, PR China
[b] Department of Electronics, Information and Bioengineering, Politecnico di Milano, Milan, 20133, Italy

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ABSTRACT

The problem of task-space tracking control of a free-floating space robot is addressed in this paper. Based on a fixed-time extended state observer (FXESO), a fixed-time position and attitude control (FXPAC) method is developed. The attitude of the end-effector is represented in modified Rodrigues parameters (MRPs). The FXESO provides the estimation of the joint velocity and the lumped disturbance for the control system. Based on the backstepping technique and a power integrator, the FXPAC method is designed. This control scheme can guarantee tracking errors converging to a neighborhood of the origin within a fixed time in the presence of the external disturbance. Moreover, since the inverse of the generalized Jacobian matrix is involved in the controller, it is necessary to consider dynamic singularities. To handle these singularities, a method of combining the singularity separation and damped reciprocal (SSDR) is applied to the space robot. Numerical simulations are presented to demonstrate the effectiveness of the proposed method.

1. Introduction

Free-floating space robots have been widely utilized in space explorations, such as maintenance and construction of space stations, and capturing tumbling satellites. Tracking control is particularly important for applications of space robots. During the last decades, there have been various robot control methods. Since the robot motor directly drives the joint, there have been a lot of results about joint-space controllers [1–3]. However, some missions are defined in task space, such as Cartesian space or sensor/image space. To ensure asymptotic end-effector position and orientation tracking, a full-state feedback controller and an output feedback controller were designed for robot manipulators [4]. For task-space tracking control of redundant robotic systems, an online kernel-based learning approach and a non-linear model predictive control with obstacle avoidance were developed in [5] and [6], respectively. For free-flying space robots, [7] and [8] proposed a fuzzy logic system adaptive controller and an adaptive sliding mode controller in task space, respectively.

Compared with the asymptotic control laws, finite-time control can shorten the convergence time, improve control accuracies, and reject the disturbance better. This kind of methods has been also applied to the robotic systems. In terms of tracking control in joint space, finite-time control was respectively combined with neural network [9], fuzzy strategy [10], and sliding mode control [11]. With regard to tracking control in task space, a sliding mode controller with finite-time stability was proposed for free-flying space robots [12]. Based on the derivative of the dynamic equation and a new non-singular terminal sliding manifold, finite-time trajectory tracking control in task space was developed for robotic manipulators [13]. Considering vibration suppression, finite-time control based on virtual control force was proposed to track a planar trajectory for a flexible space robot system [14].

There is a significant drawback of the finite-time control: the settling time depends on the initial conditions, heavily. Fixed-time control is an effective solution to this issue, which can guarantee that the settling time is bounded by a positive constant [15]. Fixed-time controllers have been applied to the attitude control of spacecraft [16,17], reusable launch vehicles [18], and the aircraft fighter control [19]. There have also been some research results about fixed-time control of robotic systems. For robots with model uncertainties, fixed-time sliding mode control and robust approximate fixed-time control were respectively proposed in [20] and [21]. Considering uncertainties and input dead zone together, fixed-time control based on the neural network was presented [22]. The aforementioned control methods were mainly developed to track desired trajectories in joint space. There have been fewer results about fixed-time control in the task space for space robots. A robust task-space fault-tolerant control, which could provide global fixed-time stability, was designed [23]. Motivated by [4,16,23], fixed-time position and attitude control (FXPAC) is studied for a free-floating space robot in this paper.

Another potential problem in the design of the control system is that the velocity measurements can be contaminated by noises, which

* Corresponding author.
E-mail addresses: rongyujin@hit.edu.cn (R. Jin), paolo.rocco@polimi.it (P. Rocco), gengyh@hit.edu.cn (Y. Geng).

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may lead to poor control performance or even instability problems. An effective solution is to estimate the joint velocity using a state observer. A sliding observer was proposed to provide the joint velocity required in the robot control [24]. For double integrator systems with disturbance, [25] designed a fixed-time observer based on the uniform robust exact differentiator. Moreover, the control performance is also impacted by the model uncertainties and the disturbance. To address the problem, some researchers have focused on the extended state observer (ESO). The core of ESO is to treat the disturbance as an extended system state and design a state observer. Finite-time extended state observers (FESOs) were employed to obtain the disturbance estimation for different objects, such as spacecraft [26,27] and mechanical systems [28]. In [27,28], both the velocity and the disturbance could be estimated by FESO. Based on the homogeneous theory, fixed-time extended observers (FXESOs) were presented in [18,19,29]. In [19], the switch function involved in the traditional FXESO design was omitted based on Theorem 2 in [30]. Based on the above, a significant advantage of ESO is that the observer can provide the state of the original system and the disturbance together [18,19,27,28]. Different from FESO, the observer errors of FXESO can converge to the origin within a fixed time independent of initial conditions. Therefore, to estimate the joint velocity and the disturbance, an FXESO is designed for free-floating space robots in this paper.

In terms of the tracking control in task space for space robots, the inverse of the generalized Jacobian matrix usually needs to be computed [31,32]. Therefore, it is necessary to consider the dynamic singularities. In [13], by introducing an auxiliary matrix, the inverse of the Jacobian matrix was not employed in the controller, which meant the singularities were avoided. However, if the range of the eigenvalues of the inertia matrix is too large, the required control inputs will also become much large. To deal with singularities, the damped least-squares (DLS) method has been utilized in a robotic system with a fixed base [33,34] and in a free-floating space robot system [31]. With regard to DLS, the estimation of the minimum singularity of the Jacobian matrix should be obtained. For space robots with wrist-partitioned manipulators, a method of combining the singularity separation and damped reciprocal (SSDR) was developed [32]. With the SSDR method, the minimum singularity of the Jacobian matrix is not required and the joint velocity can be calculated directly. In this paper, inspired by [32], SSDR is introduced to handle the dynamic singularities during the control design.

This paper focuses on the position and attitude control in the task space for a free-floating space robot. The modified Rodrigues parameters (MRPs) are employed to represent the attitude of the end-effector. On the other hand, besides the advantages of finite-time control, the settling time of the fixed-time control is independent of the initial conditions. Thus, a novel fixed-time control is developed to track the desired position and attitude trajectories for space robots. In the presence of the unknown joint velocity and disturbance, an FXESO is designed to provide these information for the system. Moreover, the SSDR method is employed to deal with dynamic singularities. Therefore, considering dynamic singularities, a novel FXPAC based on an FXESO is studied in this paper. The main contributions of this paper are summarized as follows:

1. A novel FXPAC method is proposed for a free-floating space robot. In the presence of the disturbance, the fixed-time controller is designed based on the backstepping technique and a power integrator.
2. A fixed-time extended state observer (FXESO) based on homogeneous theory is employed to obtain the estimation of the joint velocity and the lumped disturbance. Different from [18] and [19], in the proof, the derivative of the lumped disturbance is considered in the simplified systems.
3. The SSDR method is introduced to handle dynamic singularities for a space robot. The joint velocity in the kinematic equation can be calculated directly and the estimation of the minimum singularity of the Jacobian matrix is not required. Therefore, the computation burden of SSDR is less than that of the traditional DLS method.

The rest of this paper is organized as follows. In Section 2, the kinematic and dynamic equations of free-floating space robots are given. Considering dynamic singularities, a novel FXPAC method based on an FXESO is proposed in Section 3. In Section 4, numerical simulations are presented. Finally, Section 5 concludes this paper.

2. Preliminaries and problem statement

2.1. Notations, definitions and lemmas

For an arbitrary vector $\boldsymbol{v}$, $\boldsymbol{v}^{\times}$ is a skew symmetric matrix defined as $\boldsymbol{v}^{\times} = \begin{bmatrix} 0 & -v_3 & v_2; v_3 & 0 & -v_1; -v_2 & v_1 & 0 \end{bmatrix}$. $\boldsymbol{E}_n$ denotes an identity matrix with size $n \times n$. The symbol $\|\cdot\|$ represents the Euclidean norm of vectors or the induced norm for matrices. For arbitrary vector $\boldsymbol{x} = \begin{bmatrix} x_1, x_2, \ldots, x_n \end{bmatrix}^{\mathrm{T}}$, $\mathrm{sig}^r(\boldsymbol{x}) = \begin{bmatrix} |x_1|^r \mathrm{sign}(x_1), |x_2|^r \mathrm{sign}(x_2), \ldots, |x_n|^r \mathrm{sign}(x_n) \end{bmatrix}^{\mathrm{T}}$ and $\mathrm{sign}(\boldsymbol{x}) = \begin{bmatrix} \mathrm{sign}(x_1), \mathrm{sign}(x_2), \cdots, \mathrm{sign}(x_n) \end{bmatrix}^{\mathrm{T}}$, where $\mathrm{sign}(\cdot)$ denotes the sign function. For a symmetric matrix $\boldsymbol{N} \in \boldsymbol{R}^{n\times n}$, $\lambda_{\min}\{\boldsymbol{N}\}$ and $\lambda_{\max}\{\boldsymbol{N}\}$ represent the minimum and maximum eigenvalues of $\boldsymbol{N}$, respectively.

Consider the following system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}(t)),\ \boldsymbol{x}(0) = \boldsymbol{x}_0, f(0) = 0,\ \boldsymbol{x} \in \Omega \subset \boldsymbol{R}^n \tag{1}$$

where $f : \Omega \times \boldsymbol{R}^+ \to \boldsymbol{R}^n$ is a continuous function in an open neighborhood Ω of the origin $\boldsymbol{x} = 0$.

Definition 1 ([18]). The origin of the system (1) is (locally) finite-time stable if it is Lyapunov stable and there is a settling time function $T(\boldsymbol{x}_0) > 0$ satisfying $\lim_{t\to t_0+T} \boldsymbol{x}(t, \boldsymbol{x}_0) = 0$ for any initial condition $\boldsymbol{x}_0 \in \Omega$ at t_0.

Definition 2 ([15]). The system (1) is termed fixed-time stable if it is finite-time stable and the settling time function $T(\boldsymbol{x}_0)$ is bounded by a positive number $T_{\max}$, that is $T(\boldsymbol{x}_0) \le T_{\max}$.

Lemma 1 ([27]). *Considering the system (1), suppose that there exists a continuous definite positive Lyapunov function $\boldsymbol{V}(\boldsymbol{x}(t)) : \boldsymbol{R}^n \to \boldsymbol{R}$ such that*

$$\dot{\boldsymbol{V}}(\boldsymbol{x}(t)) \le -a\boldsymbol{V}^b(\boldsymbol{x}(t))$$

where $a > 0$ and $0 < b < 1$. Then the system is finite-time stable with the settling time T satisfying

$$T \le \frac{1}{a(1-b)} |\boldsymbol{V}(\boldsymbol{x}_0)|^{1-b}$$

Lemma 2 ([35]). *It is assumed that continuous real-valued functions $V_1(\boldsymbol{x})$ and $V_2(\boldsymbol{x})$ are homogeneous with respect to a generalized weight υ of degrees $l_1 > 0$ and $l_2 > 0$, respectively (see [18,29,35] for more details about homogeneous theory). If $V_1(\boldsymbol{x})$ is positive definite, the following inequality holds for every $\boldsymbol{x} \in \boldsymbol{R}^n$*

$$\left[\min_{\{z:V_1(z)=1\}} V_2(\boldsymbol{z})\right] \left[V_1(\boldsymbol{x})\right]^{\frac{l_2}{l_1}} \le V_2(\boldsymbol{x}) \le \left[\max_{z:V_1(z)=1} V_2(\boldsymbol{z})\right] \left[V_1(\boldsymbol{x})\right]^{\frac{l_2}{l_1}} \tag{2}$$

Lemma 3 ([18]). *For the system (1), suppose that there exists a Lyapunov function $V(\boldsymbol{x}(t)) : \boldsymbol{R}^n \to \boldsymbol{R}$ satisfying*

$$\dot{V}(\boldsymbol{x}(t)) \le -\left(\rho_1 V^{c_1}(\boldsymbol{x}(t)) + \rho_2 V^{c_2}(\boldsymbol{x}(t))\right)$$

where ρ_1, ρ_2, $c_1 < 1$, and $c_2 > 1$ are positive constants. Then the system (1) is fixed-time stable, and the settling time is bounded by

$$T \le \frac{1}{\rho_1(1-c_1)} + \frac{1}{\rho_2(c_2-1)}$$

regardless of the initial conditions.

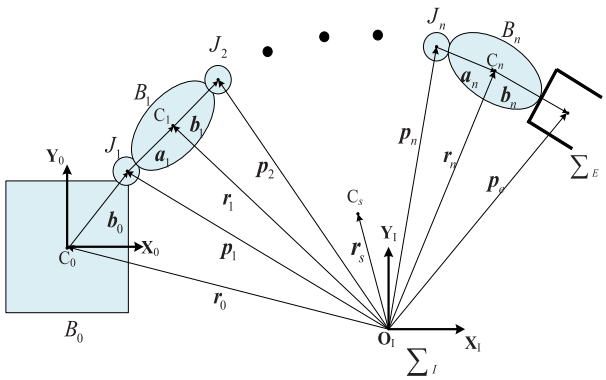


Fig. 1. General model of a space robotic system.

Table 1
The space robot symbols used in the paper.

Symbols	Representation
B_0, B_i	The base body and the ith link of the manipulator ($i = 1, 2, \dots, n$)
C_i, C_s	Mass centers of B_i and the system
$\Sigma_I, \Sigma_E, \Sigma_i$	The inertia frame, the end-effector frame, and the fixed frame of B_i
$\boldsymbol{a}_i, \boldsymbol{b}_i$	Position vectors from the origin of Σ_i to C_i and from C_i to the origin of Σ_{i+1}
$\boldsymbol{r}_i, \boldsymbol{r}_s$	Position vectors of C_i and C_s
$\boldsymbol{p}_i, \boldsymbol{p}_e$	Position vectors of the origins of Σ_i and Σ_E
$\boldsymbol{\omega}_0, \boldsymbol{\omega}_e$	Angular velocities of the base and the end-effector
$\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_n]^{\mathrm{T}}$	Joint angle of the manipulator
$(\cdot)^k$	Vector $(\cdot)$ expressed in the frame $\Sigma_k (k = 1, 2, \dots, n, E, I)$; if Σ_k is the inertia frame, the superscript k can be missed.

Lemma 4 ([16]). *For any real numbers x, y, and $0 < c \le 1$, the following inequality holds.*

$$|sig^c(x) - sig^c(y)| \le 2^{1-c}|x - y|^c$$

Lemma 5 ([17]). *Let real numbers $x_1, x_2, \dots, x_s \ge 0$, $c > 1$ and $0 < b \le 1$. Then the following inequalities hold.*

$$s^{1-c}\left(\sum_{i=1}^{s} x_i\right)^c \le \sum_{i=1}^{s} x_i^c \le \left(\sum_{i=1}^{s} x_i\right)^c$$
$$\left(\sum_{i=1}^{s} x_i\right)^b \le \sum_{i=1}^{s} x_i^b \le s^{1-b}\left(\sum_{i=1}^{s} x_i\right)^b$$

Lemma 6 ([36]). *Defining real numbers x, y, $c_1 > 0$, $c_2 > 0$, and $\gamma > 0$, we can obtain*

$$|x|^{c_1}|y|^{c_2} \le \frac{c_1\gamma|x|^{c_1+c_2}}{c_1+c_2} + \frac{c_2\gamma^{-\frac{c_1}{c_2}}|y|^{c_1+c_2}}{c_1+c_2}$$

2.2. Kinematics and dynamics of free-floating space robots

A space robot is composed of a spacecraft (base) and a manipulator with n degrees of freedom (DOF) as shown in Fig. 1. The symbols related to the space robot are listed in Table 1.

The dynamic equation of free-floating space robots with external disturbance can be expressed as

$$\boldsymbol{H}_g(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \boldsymbol{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} = \boldsymbol{\tau} + \boldsymbol{\tau}_d \tag{3}$$

where $\boldsymbol{H}_g \in \boldsymbol{R}^{n\times n}$ is a symmetric positive definite inertia matrix and $\boldsymbol{C}\dot{\boldsymbol{\theta}} \in \boldsymbol{R}^{n\times 1}$ denotes Coriolis/centrifugal force. $\boldsymbol{\tau} \in \boldsymbol{R}^{n\times 1}$ and $\boldsymbol{\tau}_d \in \boldsymbol{R}^{n\times 1}$ are respectively control input and external disturbance.

Define the velocities of the end-effector and the base as $\dot{\boldsymbol{x}}_e = \left[\dot{\boldsymbol{p}}_e^{\mathrm{T}}\ \boldsymbol{\omega}_e^{\mathrm{T}}\right]^{\mathrm{T}}$ and $\dot{\boldsymbol{x}}_0 = \left[\dot{\boldsymbol{r}}_0^{\mathrm{T}}\ \boldsymbol{\omega}_0^{\mathrm{T}}\right]^{\mathrm{T}}$, respectively. Supposing that the initial momentum of the space robot system is zero, the differential mappings between $\boldsymbol{\theta}$ and $\boldsymbol{x}_e$ can be given as

$$\dot{\boldsymbol{x}}_e = \boldsymbol{J}_b\dot{\boldsymbol{x}}_0 + \boldsymbol{J}_m\dot{\boldsymbol{\theta}} = \boldsymbol{J}_g\dot{\boldsymbol{\theta}} \tag{4}$$

where $\boldsymbol{J}_b$ and $\boldsymbol{J}_m$ are Jacobian matrices of the base and the manipulator, and $\boldsymbol{J}_g$ is termed the generalized Jacobian matrix, obtained in [37].

In this paper, the attitude of the end-effector represented with the MRPs is defined by $\boldsymbol{q}_e = \boldsymbol{r}\tan\left(\frac{\psi(t)}{4}\right)$, where $\boldsymbol{r}$ is the unit rotation axis and $\psi \in [0, 2\pi)$ is the rotation angle around $\boldsymbol{r}$. The time derivative of $\boldsymbol{q}_e$ can be expressed by the angular velocity $\boldsymbol{\omega}_e$ as follows:

$$\dot{\boldsymbol{q}}_e = \boldsymbol{G}(\boldsymbol{q}_e)\boldsymbol{\omega}_e^E \tag{5}$$

where $\boldsymbol{\omega}_e^E$ represents the vector $\boldsymbol{\omega}_e$ expressed in the end-effector frame Σ_E and $\boldsymbol{G}(\boldsymbol{q}_e)$ is denoted as

$$\boldsymbol{G}(\boldsymbol{q}_e) = \frac{1}{2}\left[\frac{1 - \boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{q}_e}{2}\boldsymbol{E}_3 + \boldsymbol{q}_e^{\times} + \boldsymbol{q}_e\boldsymbol{q}_e^{\mathrm{T}}\right]$$

The following two properties are valid:

$$\boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{G}(\boldsymbol{q}_e)\boldsymbol{\omega}_e^E = \frac{1 + \boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{q}_e}{4}\boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{\omega}_e^E$$
$$\boldsymbol{G}(\boldsymbol{q}_e)\boldsymbol{G}^{\mathrm{T}}(\boldsymbol{q}_e) = \left(\frac{1 + \boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{q}_e}{4}\right)^2\boldsymbol{E}_3$$

$\boldsymbol{p}_d$, $\boldsymbol{q}_d$, and $\boldsymbol{\omega}_d$ are denoted as the desired end-effector trajectories of the position, the attitude, and the angular velocity, respectively. Then, the position error $\boldsymbol{e}_p$ and the attitude error $\boldsymbol{e}_q$ are defined as

$$\boldsymbol{e}_p = \boldsymbol{p}_e - \boldsymbol{p}_d$$
$$\boldsymbol{e}_q = \frac{\boldsymbol{q}_d\left(\boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{q}_e - 1\right) + \boldsymbol{q}_e\left(1 - \boldsymbol{q}_d^{\mathrm{T}}\boldsymbol{q}_d\right) - 2\boldsymbol{q}_d^{\times}\boldsymbol{q}_e}{1 + \boldsymbol{q}_d^{\mathrm{T}}\boldsymbol{q}_d\boldsymbol{q}_e^{\mathrm{T}}\boldsymbol{q}_e + 2\boldsymbol{q}_d^{\mathrm{T}}\boldsymbol{q}_e}$$

Similar to Eq. (5), the differential of $\boldsymbol{e}_q$ can be derived as

$$\dot{\boldsymbol{e}}_q = \boldsymbol{G}(\boldsymbol{e}_q)\boldsymbol{e}_\omega^E \tag{6}$$

where $\boldsymbol{e}_\omega = \boldsymbol{\omega}_e - \boldsymbol{\omega}_d$. Define $\dot{\boldsymbol{x}}_e' = \left[\boldsymbol{p}_e^{\mathrm{T}}\ (\boldsymbol{\omega}_e^E)^{\mathrm{T}}\right]^{\mathrm{T}}$ and $\dot{\boldsymbol{x}}_d' = \left[\boldsymbol{p}_d^{\mathrm{T}}\ (\boldsymbol{\omega}_d^E)^{\mathrm{T}}\right]^{\mathrm{T}}$. Then, the state error $\boldsymbol{e}_1$ and the velocity error $\boldsymbol{e}_2$ of the end-effector can be defined as

$$\boldsymbol{e}_1 = \begin{bmatrix} \boldsymbol{e}_p \\ \boldsymbol{e}_q \end{bmatrix} \quad \boldsymbol{e}_2 = \begin{bmatrix} \dot{\boldsymbol{e}}_p \\ \boldsymbol{e}_\omega^E \end{bmatrix} = \dot{\boldsymbol{x}}_e' - \dot{\boldsymbol{x}}_d' \tag{7}$$

Based on (3), (4), (6), and (7), after some simple operations, the time derivatives of the tracking errors can be expressed as

$$\dot{\boldsymbol{e}}_1 = \boldsymbol{J}_e\boldsymbol{e}_2 \tag{8}$$
$$\dot{\boldsymbol{e}}_2 = \boldsymbol{J}_{Eg}\boldsymbol{H}_g^{-1}(\boldsymbol{\tau} - \boldsymbol{C}\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_d) + \dot{\boldsymbol{J}}_{Eg}\dot{\boldsymbol{\theta}} - \ddot{\boldsymbol{x}}_d' \tag{9}$$

where $\boldsymbol{J}_{Eg} = \boldsymbol{J}_E\boldsymbol{J}_g$ and

$$\boldsymbol{J}_e = \begin{bmatrix} \boldsymbol{E}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{G}(\boldsymbol{e}_q) \end{bmatrix} \quad \boldsymbol{J}_E = \begin{bmatrix} \boldsymbol{E}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{R}_E^{\mathrm{T}} \end{bmatrix}$$

$\boldsymbol{R}_E \in \boldsymbol{R}^{3\times 3}$ is the rotation matrix of the end-effector frame relative to the inertial frame.

In the premise of unknown joint velocity, a control law $\boldsymbol{\tau}$ is designed to achieve the objective that the tracking errors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ converge to a small bounded region in a fixed time independent of initial conditions.

Remark 1. It is much more convenient to establish the relationship in the end-effector frame between the attitude error $\boldsymbol{e}_q$ and the angular velocity error $\boldsymbol{e}_\omega$. Therefore, Eq. (5) is represented in the end-effector frame instead of the inertial frame.

3. Observer-based fixed-time tracking control in task space

3.1. Fixed-time extended state observer (FXESO)

In this section, FXESO is designed to obtain the estimation of the joint velocity and the disturbance.

Define $\boldsymbol{x}_1 = \boldsymbol{\theta}$, $\boldsymbol{x}_2 = \dot{\boldsymbol{\theta}}$, and $\boldsymbol{x}_3 = \boldsymbol{\tau}_l = \boldsymbol{H}_g^{-1}(-\boldsymbol{C}\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_d)$, respectively. According to the dynamic equation, a reconstructed system is obtained as follows.

$$\begin{cases} \dot{\boldsymbol{x}}_1 = \boldsymbol{x}_2 \\ \dot{\boldsymbol{x}}_2 = \boldsymbol{x}_3 + \boldsymbol{H}_g^{-1}\boldsymbol{u} \\ \dot{\boldsymbol{x}}_3 = \dot{\boldsymbol{\tau}}_l \end{cases} \tag{10}$$

Assumption 1. The external disturbance $\boldsymbol{\tau}_d$, $\boldsymbol{H}_g$, $\boldsymbol{C}$ and their derivatives are bounded [28], which means that the lumped disturbance $\boldsymbol{\tau}_l$ and its derivative $\dot{\boldsymbol{\tau}}_l$ are also bounded. $|\dot{\tau}_{li}| \le \overline{\tau}_{li}\,(i = 1, 2, \ldots, n)$ are satisfied, where $\overline{\tau}_{li}\,(i = 1, 2, \ldots, n)$ are positive constants.

The essential characteristic of ESO is that the lumped disturbance $\boldsymbol{\tau}_l$ is regarded as a system state. Therefore, both the joint velocity $\dot{\boldsymbol{\theta}}$ and $\boldsymbol{\tau}_l$ can be estimated by the state observer.

Let $\hat{\boldsymbol{x}}_i\,(i = 1, 2, 3)$ be the estimates of $\boldsymbol{x}_i$. The observer errors are defined as $\tilde{\boldsymbol{x}}_i = \hat{\boldsymbol{x}}_i - \boldsymbol{x}_i$. An ESO is designed as follows

$$\begin{cases} \dot{\hat{\boldsymbol{x}}}_1 = \hat{\boldsymbol{x}}_2 - k_{o1}\Lambda_s \mathrm{sig}^{p_1}(\tilde{\boldsymbol{x}}_1) - k_{o1}(1-\Lambda_s)\mathrm{sig}^{q_1}(\tilde{\boldsymbol{x}}_1) \\ \dot{\hat{\boldsymbol{x}}}_2 = \hat{\boldsymbol{x}}_3 + \boldsymbol{H}_g^{-1}\boldsymbol{u} - k_{o2}\Lambda_s \mathrm{sig}^{p_2}(\tilde{\boldsymbol{x}}_1) - k_{o2}(1-\Lambda_s)\mathrm{sig}^{q_2}(\tilde{\boldsymbol{x}}_1) \\ \dot{\hat{\boldsymbol{x}}}_3 = -k_{o3}\Lambda_s \mathrm{sig}^{p_3}(\tilde{\boldsymbol{x}}_1) - k_{o3}(1-\Lambda_s)\mathrm{sig}^{q_3}(\tilde{\boldsymbol{x}}_1) \end{cases} \tag{11}$$

where k_{o1}, k_{o2}, and k_{o3} are observer gains. $p_i = ip - (i-1)$ and $q_i = iq - (i-1), i = 1, 2, 3$ are satisfied, where $p \in (1-\varepsilon_1, 1)$ and $q \in (1, 1+\varepsilon_2)$ are two positive constants. Both ε_1 and ε_2 are two sufficiently small positive numbers. In addition, $\Lambda_s : [0, \infty) \to \{0, 1\}$ is a switch function as follows:

$$\Lambda_s = \begin{cases} 0 & t \le T_s \\ 1 & t > T_s \end{cases} \tag{12}$$

where T_s is termed the switching time. Then, the estimation error of the observer can be represented as

$$\begin{cases} \dot{\tilde{\boldsymbol{x}}}_1 = \tilde{\boldsymbol{x}}_2 - k_{o1}\Lambda_s \mathrm{sig}^{p_1}(\tilde{\boldsymbol{x}}_1) - k_{o1}(1-\Lambda_s)\mathrm{sig}^{q_1}(\tilde{\boldsymbol{x}}_1) \\ \dot{\tilde{\boldsymbol{x}}}_2 = \tilde{\boldsymbol{x}}_3 - k_{o2}\Lambda_s \mathrm{sig}^{p_2}(\tilde{\boldsymbol{x}}_1) - k_{o2}(1-\Lambda_s)\mathrm{sig}^{q_2}(\tilde{\boldsymbol{x}}_1) \\ \dot{\tilde{\boldsymbol{x}}}_3 = -k_{o3}\Lambda_s \mathrm{sig}^{p_3}(\tilde{\boldsymbol{x}}_1) - k_{o3}(1-\Lambda_s)\mathrm{sig}^{q_3}(\tilde{\boldsymbol{x}}_1) - \dot{\boldsymbol{\tau}}_l \end{cases} \tag{13}$$

Theorem 1. *Considering the observer (11) for the system (10) and Assumption 1, the observer errors converge to a bounded set in fixed time independent of the initial conditions, if the observer parameters satisfy $k_{oi} > 0\,(i = 1, 2, 3)$, $k_{o1}k_{o2} > k_{o3}$, $\varepsilon_1 \le \frac{1}{3}$, and $\varepsilon_2 \le \frac{1}{2}$.*

Proof. If $k_{oi} > 0\,(i = 1, 2, 3)$ and $k_{o1}k_{o2} > k_{o3}$ are satisfied, the following matrix is a Hurwitz matrix.

$$\boldsymbol{A} = \begin{bmatrix} -k_{o1} & 1 & 0 \\ -k_{o2} & 0 & 1 \\ -k_{o3} & 0 & 0 \end{bmatrix}$$

Therefore, there exist two symmetric positive definite matrices $\boldsymbol{P}$ and $\boldsymbol{Q}$ such that $\boldsymbol{P}\boldsymbol{A} + \boldsymbol{A}^{\mathrm{T}}\boldsymbol{P} = -\boldsymbol{Q}$.

If $t \le Ts$, the error system (13) can be rewritten as

$$\begin{cases} \dot{\boldsymbol{z}}_1 = \boldsymbol{z}_2 - k_{o1}\mathrm{sig}^{q_1}(\boldsymbol{z}_1) \\ \dot{\boldsymbol{z}}_2 = \boldsymbol{z}_3 - k_{o2}\mathrm{sig}^{q_2}(\boldsymbol{z}_1) \\ \dot{\boldsymbol{z}}_3 = -k_{o3}\mathrm{sig}^{q_3}(\boldsymbol{z}_1) - \dot{\boldsymbol{\tau}}_l \end{cases} \tag{14}$$

A Lyapunov function candidate is selected as

$$V_1(\boldsymbol{z}, q) = \sum_{i=1}^{n} \boldsymbol{\Xi}_i^{\mathrm{T}}(\boldsymbol{z}, q)\boldsymbol{P}\boldsymbol{\Xi}_i(\boldsymbol{z}, q) \tag{15}$$

where $\boldsymbol{\Xi}_i(\boldsymbol{z}, q) = \begin{bmatrix} z_{1i} & z_{2i}^{\frac{1}{q_1}} & z_{3i}^{\frac{1}{q_2}} \end{bmatrix}^{\mathrm{T}}$. The time derivative of $V_1(\boldsymbol{z}, p)$ can be divided into two parts:

$$\dot{V}_1(\boldsymbol{z}, q) = V_{11}'(\boldsymbol{z}, q) + V_{12}'(\boldsymbol{z}, q) \tag{16}$$

$V_{11}'(\boldsymbol{z}, q)$ can be considered as the simplified Lyapunov function $\dot{V}_1(\boldsymbol{z}, q)$ with $\dot{\boldsymbol{\tau}}_l = \boldsymbol{0}$. $V_{12}'(\boldsymbol{z}, q)$ is the part induced by $\dot{\boldsymbol{\tau}}_l$ and derived by

$$V_{12}'(\boldsymbol{z}, q) = \sum_{i=1}^{n} \dot{\tau}_{li} f_i(\boldsymbol{z}, q) \le \overline{\tau}_{l\max}|V_{13}'(\boldsymbol{z}, q)| \tag{17}$$

where $\overline{\tau}_{l\max} = \max\{\overline{\tau}_{li}, i = 1, 2, \ldots, n\}$, $V_{13}'(\boldsymbol{z}, q) = \sum_{i=1}^{n} f_i(\boldsymbol{z}, q)$, and $f_i(\boldsymbol{z}, q) = -\frac{2}{q_2} z_3^{\frac{1}{q_2}-1} \boldsymbol{\Xi}_i^{\mathrm{T}}(\boldsymbol{z}, q)\boldsymbol{P}_3$. $\boldsymbol{P}_3$ is the third column of the matrix $\boldsymbol{P}$.

According to Assumption 1 and [28], if $q = 1$ holds, the system (14) is finite-time stable. $V_{11}'(\boldsymbol{z}, 1)$ and $V_{13}'(\boldsymbol{z}, 1)$ can be calculated as

$$\begin{aligned} V_{11}'(\boldsymbol{z}, 1) &= -\textstyle\sum_{i=1}^{n} \boldsymbol{\Xi}_i^{\mathrm{T}}(\boldsymbol{z}, 1)\boldsymbol{Q}\boldsymbol{\Xi}_i(\boldsymbol{z}, 1) \le -\frac{\lambda_{\min}\{\boldsymbol{Q}\}}{\lambda_{\max}\{\boldsymbol{P}\}} V_1(\boldsymbol{z}, 1) \\ V_{13}'(\boldsymbol{z}, 1) &= -\textstyle\sum_{i=1}^{n} 2\boldsymbol{\Xi}_i^{\mathrm{T}}(\boldsymbol{z}, 1)\boldsymbol{P}_3 \le \frac{2\|\boldsymbol{P}_3\|}{\sqrt{\lambda_{\min}\{\boldsymbol{P}\}}} V_1^{\frac{1}{2}}(\boldsymbol{z}, 1) \end{aligned} \tag{18}$$

Note that $V_{11}'(\boldsymbol{z}, q)$ is continuous with respect to its arguments q and z. $V_{11}'(\boldsymbol{z}, 1) \le 0$ implies that there exists a small constant ε_2 such that $V_{11}'(\boldsymbol{z}, q) \le 0$ for $q \in (1, 1+\varepsilon_2)$ [38].

With respect to the weights $r_j = (j-1)q - (j-2)\ (j = 1, 2, 3)$, $V_1(\boldsymbol{z}, q)$, $V_{11}'(\boldsymbol{z}, q)$, and $V_{13}'(\boldsymbol{z}, q)$ are homogeneous of degrees 2, $q+1$, and $3-2q$, respectively. $3-2q > 0$ is guaranteed by $\varepsilon_2 < \frac{1}{2}$. Based on Lemma 2 and $V_{11}'(\boldsymbol{z}, q) \le 0$, the following inequalities are obtained:

$$\begin{aligned} V_{11}'(\boldsymbol{z}, q) &\le -\kappa_1(\boldsymbol{z}, q) V_1^{\frac{q+1}{2}}(\boldsymbol{z}, q) \\ V_{13}'(\boldsymbol{z}, q) &\le \kappa_2(\boldsymbol{z}, q) V_1^{\frac{3-2q}{2}}(\boldsymbol{z}, q) \end{aligned} \tag{19}$$

where $\lim_{q\to 1}\kappa_1(\boldsymbol{z}, q) = \frac{\lambda_{\min}\{\boldsymbol{Q}\}}{\lambda_{\max}\{\boldsymbol{P}\}}$ and $\lim_{q\to 1}\kappa_2(\boldsymbol{z}, q) = \frac{2\|\boldsymbol{P}_3\|}{\sqrt{\lambda_{\min}\{\boldsymbol{P}\}}}$. Since the right side of the system (14) is continuous with respect to q, there exists a small constant ε_2 such that for $q \in (1, 1+\varepsilon_2)$, $\kappa_1(\boldsymbol{z}, q) \ge \frac{\lambda_{\min}\{\boldsymbol{Q}\}}{2\lambda_{\max}\{\boldsymbol{P}\}} = \overline{\kappa}_1$ and $\kappa_2(\boldsymbol{z}, q) \le \frac{4\|\boldsymbol{P}_3\|}{\sqrt{\lambda_{\min}\{\boldsymbol{P}\}}} = \overline{\kappa}_2$. Based on (16), (17), and (19), we can obtain

$$\begin{aligned} \dot{V}_1(\boldsymbol{z}, q) &\le -\overline{\kappa}_1 V_1^{\frac{q+1}{2}}(\boldsymbol{z}, q) + \overline{\kappa}_2\overline{\tau}_{l\max} V_1^{\frac{3-2q}{2}}(\boldsymbol{z}, q) \\ &\le -\left(\frac{1}{2}\overline{\kappa}_1 V_1^{\frac{3q-2}{2}}(\boldsymbol{z}, q) - \overline{\kappa}_2\overline{\tau}_{l\max}\right) V_1^{\frac{3-2q}{2}}(\boldsymbol{z}, q) - \frac{1}{2}\overline{\kappa}_1 V_1^{\frac{q+1}{2}}(\boldsymbol{z}, q) \end{aligned}$$

If $V_1(\boldsymbol{z}, q) \ge (\frac{2\overline{\kappa}_2\overline{\tau}_{l\max}}{\overline{\kappa}_1})^{\frac{2}{3q-2}}$, then $\dot{V}_1(\boldsymbol{z}, q) \le -\frac{1}{2}\overline{\kappa}_1 V_1^{\frac{q+1}{2}}(\boldsymbol{z}, q)$. Therefore, when $t = T_s$, $V_1(\boldsymbol{z}, q)$ satisfies

$$\begin{aligned} V_1(\boldsymbol{z}, q)|_{T_s} &\le \left(\frac{q-1}{8}\frac{\lambda_{\min}\{\boldsymbol{Q}\}}{\lambda_{\max}\{\boldsymbol{P}\}}T_s + (V_1(\boldsymbol{z}, q)|_0)^{\frac{1-q}{2}}\right)^{\frac{2}{1-q}} \\ &\le \left(\frac{q-1}{8}\frac{\lambda_{\min}\{\boldsymbol{Q}\}}{\lambda_{\max}\{\boldsymbol{P}\}}T_s\right)^{\frac{2}{1-q}} \end{aligned}$$

$V_1(\boldsymbol{z}, q)|_{T_s}$ is finally expressed as

$$V_1(\boldsymbol{z}, q)|_{T_s} \le \overline{V}_s = \max\left\{\left(\frac{2\overline{\kappa}_2\tau_{l\max}}{\overline{\kappa}_1}\right)^{\frac{2}{3q-2}}, \left(\frac{q-1}{8}\frac{\lambda_{\min}\{\boldsymbol{Q}\}}{\lambda_{\max}\{\boldsymbol{P}\}}T_s\right)^{\frac{2}{1-q}}\right\} \tag{20}$$

In terms of $t > Ts$, consider the Lyapunov function candidate of the system (13) as $V_2(\boldsymbol{z}, p) = \sum_{i=1}^{n} \boldsymbol{\Gamma}_i^{\mathrm{T}}(\boldsymbol{z}, p)\boldsymbol{P}\boldsymbol{\Gamma}_i(\boldsymbol{z}, p)$, where $\boldsymbol{\Gamma}_i(\boldsymbol{z}, p) = \begin{bmatrix} z_{1i} & z_{2i}^{\frac{1}{p_1}} & z_{3i}^{\frac{1}{p_2}} \end{bmatrix}^{\mathrm{T}}$. Similarly, if $\varepsilon_1 \le \frac{1}{3}$ and $V_2(\boldsymbol{z}, p) \ge \left(\frac{2\overline{\kappa}_2\overline{\tau}_{l\max}}{\overline{\kappa}_1}\right)^{\frac{2}{3p-2}}$, the time derivative of the Lyapunov function satisfies

$$\dot{V}_2(\boldsymbol{z}, p) \le -\frac{\lambda_{\min}\{\boldsymbol{Q}\}}{4\lambda_{\max}\{\boldsymbol{P}\}} V_2^{\frac{p+1}{2}}(\boldsymbol{z}, p) \tag{21}$$

According to Lemma 1, since $p < 1$, the system is finite-time stable with the initial condition $V_1(\boldsymbol{z}, q)|_{T_s}$. The estimation errors of the system (13) will converge to $\Omega_1 = \left\{\boldsymbol{\Gamma}_i \in \boldsymbol{R}^3 | \|\boldsymbol{\Gamma}_i\| \le \frac{1}{\sqrt{\lambda_{\min}\{\boldsymbol{P}\}}}\left(\frac{2\overline{\kappa}_2\overline{\tau}_{l\max}}{\overline{\kappa}_1}\right)^{\left(\frac{1}{3p-2}\right)}\right\}$ in fixed time, which is bounded by $T_1 = T_s + \frac{4\lambda_{\max}\{\boldsymbol{P}\}}{\lambda_{\min}\{\boldsymbol{Q}\}}\frac{2}{1-p}\overline{V}_s^{\frac{1-p}{2}}$. □

Remark 2. In the proof of [18] and [19], $\dot{\boldsymbol{\tau}}_l$ in the simplified system (14) is ignored. On the contrary, in this paper, $\dot{\boldsymbol{\tau}}_l$ is considered, which makes the proof more rigorous.

3.2. Fixed-time position and attitude control (FXPAC)

In this section, a novel FXPAC based on the FXESO (11) is developed for space robots. The main objective is to design a control law $\boldsymbol{\tau}$ such that, in the presence of external disturbance, the tracking errors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ can converge to a neighborhood of the origin in fixed time regardless of initial conditions. Inspired by [16], a backstepping technique and a power integrator are employed. The procedure of the control design mainly includes a virtual controller design and a control law design.

Step 1: Virtual controller design

As a Lyapunov function candidate, consider

$$V_3 = \frac{1}{8}\boldsymbol{e}_p^{\mathrm{T}}\boldsymbol{e}_p + \frac{1}{2}\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q \tag{22}$$

To facilitate the controller design, an auxiliary variable is defined as $\boldsymbol{v}_a = \boldsymbol{e}_2 + k_a \mathrm{sig}^{\beta}(\boldsymbol{e}_1)$, where $k_a > 0$ and $\beta > 1$ are the control parameters. The time derivative of V_3 can be calculated as

$$\begin{aligned}\dot{V}_3 &= \frac{1}{4}\boldsymbol{e}_p^{\mathrm{T}}\dot{\boldsymbol{e}}_p + \frac{1+\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q}{4}\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_\omega^E = \boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\boldsymbol{e}_2 \\ &= \boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\boldsymbol{v}_a - k_a\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_1)\end{aligned} \tag{23}$$

where $k_v = \frac{1}{4}\mathrm{diag}(1,1,1,1+\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q, 1+\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q, 1+\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q)$. Consider $\boldsymbol{v}_a$ as a virtual control input, and design a control law for $\boldsymbol{v}_a$ as $\boldsymbol{v}_1 = -k_1\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1)$, where $k_1 > 0$, $\alpha_1 = (1+\alpha)/2$, and $0 < \alpha < 1$. Then, (23) can be written as

$$\dot{V}_3 = -k_1\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1) - k_a\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_1) + \boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v(\boldsymbol{v}_a - \boldsymbol{v}_1) \tag{24}$$

k_{vi} is the ith diagonal element of $\boldsymbol{k}_v$, and $\boldsymbol{e}_3 = \mathrm{sig}^{\frac{1}{\alpha_1}}(\boldsymbol{v}_a) - \mathrm{sig}^{\frac{1}{\alpha_1}}(\boldsymbol{v}_1)$ is defined. Based on Lemmas 4 and 6, the following inequality can be derived.

$$\begin{aligned}\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v(\boldsymbol{v}_a - \boldsymbol{v}_1) &\le \sum_{i=1}^{6} k_{vi}|e_{1i}||v_{ai} - v_{1i}| \le \sum_{i=1}^{6} 2^{1-\alpha_1}k_{vi}|e_{1i}||e_{3i}|^{\alpha_1} \\ &\le \sum_{i=1}^{6}\frac{2^{1-\alpha_1}}{1+\alpha_1}k_{vi}\left(|e_{1i}|^{1+\alpha_1} + \alpha_1|e_{3i}|^{1+\alpha_1}\right) \\ &= \frac{2^{1-\alpha_1}}{1+\alpha_1}\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1) + \alpha_1\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_3)\right)\end{aligned} \tag{25}$$

Substituting Eq. (25) into Eq. (24), $\dot{V}_3$ satisfies

$$\begin{aligned}\dot{V}_3 \le &-\left(k_1 - \frac{2^{1-\alpha_1}}{1+\alpha_1}\right)\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1) - k_a\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_1) \\ &+ \frac{2^{1-\alpha_1}}{1+\alpha_1}\alpha_1\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_3)\end{aligned} \tag{26}$$

Step 2: Control law design

The Lyapunov function candidate is selected as

$$V_5 = V_3 + V_4 = V_3 + \sum_{i=1}^{6} c_i V_{4i} \tag{27}$$

where

$$V_{4i} = \int_{v_{1i}}^{v_{ai}} \mathrm{sig}^{2-\alpha_1}\left(\mathrm{sig}^{\frac{1}{\alpha_1}}(s) - \mathrm{sig}^{\frac{1}{\alpha_1}}(v_{1i})\right)\mathrm{d}s \tag{28}$$

$$c_i = \begin{cases} \frac{1}{4} & i = 1,2,3 \\ \frac{1}{3} & i = 4,5,6 \end{cases} \tag{29}$$

V_{4i} is defined by a power integrator, which can be regarded as a Lyapunov function [36]. c_i represents the coefficient corresponding to V_{4i}. The time derivative of V_{4i} can be computed by

$$\begin{aligned}\dot{V}_{4i} =&\mathrm{sig}^{2-\alpha_1}(e_{3i})\dot{v}_{ai} + k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\dot{e}_{1i}\int_{v_{1i}}^{v_{ai}}\mathrm{sig}^{1-\alpha_1}\left(\mathrm{sig}^{\frac{1}{\alpha_1}}(s) - \mathrm{sig}^{\frac{1}{\alpha_1}}(v_{1i})\right)\mathrm{d}s \\ \le&\mathrm{sig}^{2-\alpha_1}(e_{3i})\dot{v}_{ai} + k_1^{\frac{1}{\alpha_1}}2^{1-\alpha_1}(2-\alpha_1)|e_{3i}|\dot{e}_{1i}\end{aligned} \tag{30}$$

Based on (8) and $\boldsymbol{v}_a$, $\dot{e}_{1i}$ with $i = 1,2,3$ can be calculated as

$$\dot{e}_{1i} = e_{2i} \le |e_{2i}| \le |v_{ai}| + k_a|e_{1i}|^{\beta} \tag{31}$$

Substituting Eq. (31) into Eq. (30) yields

$$\dot{V}_{4i} \le 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left[|e_{3i}||v_{ai}| + k_a|e_{3i}||e_{1i}|^{\beta}\right] + \mathrm{sig}^{2-\alpha_1}(e_{3i})\dot{v}_{ai}$$

Similar to Eq. (25), using Lemmas 4 and 6, the following inequalities can be obtained.

$$|e_{3i}||v_{ai}| \le 2^{1-\alpha_1}|e_{3i}|^{1+\alpha_1} + \frac{\alpha_1}{1+\alpha_1}|e_{1i}|^{1+\alpha_1} + \frac{k_1^{1+\alpha_1}}{1+\alpha_1}|e_{3i}|^{1+\alpha_1} \tag{32}$$

$$k_a|e_{1i}|^{\beta}|e_{3i}| \le \frac{\beta}{1+\beta}|e_{1i}|^{1+\beta} + \frac{k_a^{1+\beta}}{1+\beta}|e_{3i}|^{1+\beta} \tag{33}$$

In terms of $i = 1,2,3$, according to Eqs. (31)–(33), the range of $\dot{V}_{4i}$ can be rewritten as

$$\begin{aligned}\dot{V}_{4i} \le &m_{1p}|e_{3i}|^{1+\alpha_1} + m_{2p}|e_{3i}|^{1+\beta} + m_{3p}|e_{1i}|^{1+\alpha_1} + m_{4p}|e_{1i}|^{1+\beta} \\ &+ \mathrm{sig}^{2-\alpha_1}(e_{3i})\dot{v}_{ai}\end{aligned} \tag{34}$$

where

$$\begin{aligned} m_{1p} &= 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left(2^{1-\alpha_1} + \frac{k_1^{1+\alpha_1}}{1+\alpha_1}\right) \\ m_{2p} &= 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left(\frac{k_a^{1+\beta}}{1+\beta}\right) \\ m_{3p} &= 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left(\frac{\alpha_1}{1+\alpha_1}\right) \\ m_{4p} &= 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left(\frac{\beta}{1+\beta}\right)\end{aligned} \tag{35}$$

With regard to the attitude error $e_{1i}(i = 4,5,6)$, the following inequality is satisfied [16]:

$$\dot{e}_{1i} \le \vartheta\sum_{j=4}^{6}|v_{aj}| + k_a\vartheta\sum_{j=4}^{6}|e_{1j}|^{\beta} \tag{36}$$

where $\vartheta = \frac{1+\boldsymbol{e}_q^{\mathrm{T}}\boldsymbol{e}_q}{4}$. Similarly, $\dot{V}_{4i}(i = 4,5,6)$ can be expressed by

$$\begin{aligned}\dot{V}_{4i} \le&\mathrm{sig}^{2-\alpha_1}(e_{3i})\dot{v}_{ai} + \vartheta\left[m_{1a}|e_{3i}|^{1+\alpha_1} + m_{2a}|e_{3i}|^{1+\beta}\right. \\ &\left.+ \sum_{j=4}^{6}\left(m_{3a}|e_{1j}|^{1+\alpha_1} + m_{4a}|e_{1j}|^{1+\beta} + m_{5a}|e_{3j}|^{1+\alpha_1}\right)\right]\end{aligned} \tag{37}$$

where

$$\begin{aligned} m_{1a} &= 3\times 2^{1-\alpha_1}k_1^{\frac{1}{\alpha_1}}(2-\alpha_1)\left(\frac{2^{1-\alpha_1}}{1+\alpha_1} + \frac{k_1^{1+\alpha_1}}{1+\alpha_1}\right) \\ m_{2a} &= 3m_{2p} \\ m_{3a} &= m_{3p} \\ m_{4a} &= m_{4p} \\ m_{5a} &= m_{1p} - \tfrac{1}{3}m_{1a}\end{aligned} \tag{38}$$

Using (26), (27), (34), and (37), the time derivative of V_5 satisfies

$$\begin{aligned}\dot{V}_5 \le &-\left(k_1 - \frac{2^{1-\alpha_1}}{1+\alpha_1} - m_{3p}\right)\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1) - (k_a - m_{4p})\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_1) \\ &+ \left(2^{1-\alpha_1}\frac{\alpha_1}{1+\alpha_1} + m_{1p}\right)\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_3) + m_{2p}\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_3) \\ &+ \left[\mathrm{sig}^{2-\alpha_1}(\boldsymbol{e}_3)\right]^{\mathrm{T}}\boldsymbol{C}_\tau\dot{\boldsymbol{v}}_a\end{aligned} \tag{39}$$

where $\boldsymbol{C}_\tau = \mathrm{diag}(c_1, c_2, \ldots, c_6)$.

Since in some space robot systems, only joint angles can be measured, the joint velocity and the lumped disturbance $\boldsymbol{\tau}_l$ are provided by the observer (11) in this paper. The observer-based controller is

designed as

$$\boldsymbol{\tau} = -\boldsymbol{H}_g\hat{\boldsymbol{x}}_3 + \boldsymbol{H}_g\boldsymbol{J}_{Eg}^{-1}\Big[\ddot{\boldsymbol{x}}_d' - \dot{\hat{\boldsymbol{J}}}_{Eg}\hat{\boldsymbol{x}}_2 - k_a\beta\mathrm{diag}(|\boldsymbol{e}_1|^{\beta-1})\boldsymbol{J}_e\hat{\boldsymbol{e}}_2 \\ -(k_2\boldsymbol{C}_\tau^{-1}\boldsymbol{k}_v + k\boldsymbol{C}_\tau^{-1})\mathrm{sig}^{\alpha}(\hat{\boldsymbol{e}}_3) - (k_3\boldsymbol{C}_\tau^{-1}\boldsymbol{k}_v + k\boldsymbol{C}_\tau^{-1})\mathrm{sig}^{\alpha_1+\beta-1}(\hat{\boldsymbol{e}}_3)\Big] \quad (40)$$

where

$$\begin{aligned} &\dot{\hat{\boldsymbol{J}}}_{Eg} = \dot{\boldsymbol{J}}_{Eg}(\boldsymbol{\theta}, \hat{\boldsymbol{x}}_2) \\ &\hat{\boldsymbol{e}}_2 = \dot{\hat{\boldsymbol{x}}}_e' - \dot{\boldsymbol{x}}_d^E = \boldsymbol{e}_2 + \dot{\tilde{\boldsymbol{x}}}_e' \\ &\hat{\boldsymbol{e}}_3 = \mathrm{sig}^{\frac{1}{\alpha_1}}(\hat{\boldsymbol{v}}_a) - \mathrm{sig}^{\frac{1}{\alpha_1}}(\boldsymbol{v}_1) = \boldsymbol{e}_3 + \tilde{\boldsymbol{e}}_3 \\ &\dot{\hat{\boldsymbol{x}}}_e' = \boldsymbol{J}_{Eg}\hat{\boldsymbol{x}}_2 = \dot{\boldsymbol{x}}_e' + \dot{\tilde{\boldsymbol{x}}}_e' \\ &\hat{\boldsymbol{v}}_a = \hat{\boldsymbol{e}}_2 + k_a\mathrm{sig}^{\beta}(\boldsymbol{e}_1) = \boldsymbol{v}_a + \dot{\tilde{\boldsymbol{x}}}_e' \end{aligned} \quad (41)$$

$\dot{\tilde{\boldsymbol{x}}}_e' = \boldsymbol{J}_{Eg}\tilde{\boldsymbol{x}}_2$ and $\tilde{\boldsymbol{e}}_3 = \mathrm{sig}^{\frac{1}{\alpha_1}}(\hat{\boldsymbol{v}}_a) - \mathrm{sig}^{\frac{1}{\alpha_1}}(\boldsymbol{v}_a)$ are satisfied. With the control law (40), $\dot{\boldsymbol{v}}_a$ is derived by

$$\dot{\boldsymbol{v}}_a = -(k_2\boldsymbol{C}_\tau^{-1}\boldsymbol{k}_v + k\boldsymbol{C}_\tau^{-1})\mathrm{sig}^{\alpha}(\boldsymbol{e}_3) - (k_3\boldsymbol{C}_\tau^{-1}\boldsymbol{k}_v + k\boldsymbol{C}_\tau^{-1})\mathrm{sig}^{\alpha_1+\beta-1}(\boldsymbol{e}_3) + \boldsymbol{\Delta} \quad (42)$$

where

$$\begin{aligned} \boldsymbol{\Delta} = &-\boldsymbol{J}_{Eg}\tilde{\boldsymbol{x}}_3 + \dot{\boldsymbol{J}}_{Eg}\dot{\boldsymbol{\theta}} - \dot{\hat{\boldsymbol{J}}}_{Eg}\hat{\boldsymbol{x}}_2 - k_a\beta\mathrm{diag}(|\boldsymbol{e}_1|^{\beta-1})\boldsymbol{J}_e\dot{\tilde{\boldsymbol{x}}}_e' \\ &- (k_2\boldsymbol{A}^{-1}\boldsymbol{k}_v + k\boldsymbol{A}^{-1})\left(\mathrm{sig}^{\alpha}(\hat{\boldsymbol{e}}_3) - \mathrm{sig}^{\alpha}(\boldsymbol{e}_3)\right) \\ &- (k_3\boldsymbol{A}^{-1}\boldsymbol{k}_v + k\boldsymbol{A}^{-1})\left(\mathrm{sig}^{\alpha_1+\beta-1}(\hat{\boldsymbol{e}}_3) - \mathrm{sig}^{\alpha_1+\beta-1}(\boldsymbol{e}_3)\right) \end{aligned} \quad (43)$$

Assumption 2. Based on Theorem 1 and [13], $\tilde{\boldsymbol{x}}_2$, $\tilde{\boldsymbol{x}}_3$, $\dot{\tilde{\boldsymbol{x}}}_e^E$, $\tilde{\boldsymbol{e}}_3$, $\boldsymbol{J}_{Eg}$, and $\dot{\boldsymbol{J}}_{Eg}$ are all bounded. Therefore, $\boldsymbol{\Delta}$ is assumed to be bounded satisfying $\|\boldsymbol{\Delta}\| \le \overline{\Delta}$, where $\overline{\Delta}$ is a positive constant.

Theorem 2. *Consider the space robot system described by (3) and (4) with Assumption 2. If the observer-based controller is designed as (40), then there exist control parameters k_a, k_i $(i = 1, 2, 3)$, k, α, and β such that the tracking errors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ will converge into bounded sets within fixed time.*

Proof. Substituting (42) into (39), we can obtain

$$\begin{aligned} \dot{V}_5 \le &-\left(k_1 - \frac{2^{1-\alpha_1}}{1+\alpha_1} - m_{3p}\right)\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1) - (k_a - m_{4p})\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_1) \\ &-\left(k_2 - 2^{1-\alpha_1}\frac{\alpha_1}{1+\alpha_1} - m_{1p}\right)\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_3) - (k_3 - m_{2p})\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{k}_v\mathrm{sig}^{\beta}(\boldsymbol{e}_3) \\ &- k\boldsymbol{e}_3^{\mathrm{T}}\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_3) - k\boldsymbol{e}_3^{\mathrm{T}}\mathrm{sig}^{\beta}(\boldsymbol{e}_3) + \frac{1}{3}\overline{\Delta}\|\mathrm{sig}^{2-\alpha_1}(\boldsymbol{e}_3)\| \end{aligned} \quad (44)$$

Select the control parameters satisfying

$$\begin{aligned} &k_1 > \tfrac{2^{1-\alpha_1}}{1+\alpha_1} + m_{3p} + k_4 \\ &k_2 > 2^{1-\alpha_1}\tfrac{\alpha_1}{1+\alpha_1} + m_{1p} \\ &k_3 > m_{2p} \\ &k_a > m_{4p} + k_4 \end{aligned} \quad (45)$$

where k_4 is a positive constant. Then, using Lemma 5 and the inequality $\|\mathrm{sig}^{2-\alpha_1}(\boldsymbol{e}_3)\| \le 6^{\alpha_1/2}\|\boldsymbol{e}_3\|^{2-\alpha_1} \le 6^{\alpha_1/2}\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3\right)^{(2-\alpha_1)/2}$, $\dot{V}_5$ can be expressed as

$$\begin{aligned} \dot{V}_5 \le &-\frac{k_m}{2}\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3\right)^{\frac{1+\alpha_1}{2}} - 12^{\frac{1-\beta}{2}}k_m\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3\right)^{\frac{1+\beta}{2}} \\ &-\left[\frac{k_m}{2}\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3\right)^{\frac{\alpha}{2}} - \frac{6^{\frac{\alpha_1}{2}}}{3}\overline{\Delta}\right]\left(\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3\right)^{\frac{2-\alpha_1}{2}} \end{aligned} \quad (46)$$

where $k_m = \min\left\{\frac{k_4}{4}, k\right\}$. By using Lemma 4, $V_5 = V_3 + V_4 \le \frac{1}{2}\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \frac{2^{1-\alpha_1}}{3}\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3 \le \boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3$. Therefore, if $\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3 \ge \left(\frac{2\times 6^{\frac{\alpha_1}{2}}\overline{\Delta}}{3k_m}\right)^{\frac{2}{\alpha}}$, $\dot{V}_5$ in (46) becomes

$$\dot{V}_5 \le -\frac{k_m}{2}V_5^{\frac{1+\alpha_1}{2}} - 12^{\frac{1-\beta}{2}}k_mV_5^{\frac{1+\beta}{2}} \quad (47)$$

Based on Lemmas 4 and 5, the inequality $\sqrt{\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3} \le \left(\frac{2\times 6^{\frac{\alpha_1}{2}}\overline{\Delta}}{3k_m}\right)^{\frac{1}{\alpha}}$ implies $\|\boldsymbol{e}_1\| \le \Delta_1$ and

$$\|\boldsymbol{e}_2\| \le \|\boldsymbol{v}_a - \boldsymbol{v}_1\| + k_1\|\mathrm{sig}^{\alpha_1}(\boldsymbol{e}_1)\| + k_a\|\mathrm{sig}^{\beta}(\boldsymbol{e}_1)\| \le \Delta_2$$

where $\Delta_1 = \left(\frac{2\times 6^{\frac{\alpha_1}{2}}\overline{\Delta}}{3k_m}\right)^{\frac{1}{\alpha}}$ and $\Delta_2 = \left(2^{1-\alpha_1} + k_1\right)6^{1-\frac{\alpha_1}{2}}\Delta_1^{\alpha_1} + k_a\Delta_1^{\beta}$. Therefore, based on Lemma 3, we can conclude that $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ converge to $\Omega_2 = \left\{\boldsymbol{e}_1 \in \boldsymbol{R}^6 | \|\boldsymbol{e}_1\| \le \Delta_1\right\}$ and $\Omega_3 = \left\{\boldsymbol{e}_2 \in \boldsymbol{R}^6 | \|\boldsymbol{e}_2\| \le \Delta_2\right\}$ within fixed time $T_2 = T_1 + \frac{4}{k_m(1-\alpha_1)} + \frac{2\times 12^{\frac{\beta-1}{2}}}{k_m(\beta-1)}$. □

Remark 3. According to (31) and (36), the ranges of $\dot{e}_{1i}, i = 1, 2, 3$ (position) and $\dot{e}_{1i}, i = 4, 5, 6$ (attitude) are different. To combine the attitude part with the position part and make the calculation of $\dot{V}_4$ simple, the coefficients $c_i (i = 1, 2, \ldots, 6)$ in (29) are introduced.

3.3. Handling the dynamic singularities based on the SSDR method

If dynamic singularities occur, both $\boldsymbol{J}_g$ and $\boldsymbol{J}_{Eg}$ are not invertible. This implies that the control input $\boldsymbol{\tau}$ in (40) or the joint velocity $\dot{\boldsymbol{\theta}}$ in (4) may become too large. However, due to the mechanical limitations, the system cannot provide the required input or the joint velocity. Therefore, to address this issue, we first try to obtain the approximate angular acceleration $\ddot{\boldsymbol{\theta}}_\tau$ corresponding to $\boldsymbol{\tau}$ based on the SSDR method. Then, the control input, derived by substituting $\ddot{\boldsymbol{\theta}}_\tau$ into the dynamic equation (3), is employed instead of the original large input.

The determinant of $\boldsymbol{J}_g$ is used to determine whether dynamic singularities occur. If $|\boldsymbol{J}_g| > \varepsilon_3$ or $|\boldsymbol{J}_g| < -\varepsilon_3$, we can solve the control input according to (40) directly. If $-\varepsilon_3 \le |\boldsymbol{J}_g| \le \varepsilon_3$, the singularities should be considered.

In the case of $-\varepsilon_3 \le |\boldsymbol{J}_g| \le \varepsilon_3$, based on (9) and (40), the acceleration of the end-effector corresponding to $\boldsymbol{\tau}$ can be represented by

$$\begin{aligned} \ddot{\boldsymbol{x}}_\tau' =& \dot{\boldsymbol{e}}_2 + \ddot{\boldsymbol{x}}_d' \\ =& \ddot{\boldsymbol{x}}_d' - k_a\beta\mathrm{diag}(|\boldsymbol{e}_1|^{\beta-1})\boldsymbol{J}_e\hat{\boldsymbol{e}}_2 - (k_2\boldsymbol{A}^{-1}\boldsymbol{k}_v + k\boldsymbol{A}^{-1})\mathrm{sig}^{\alpha}(\hat{\boldsymbol{e}}_3) \\ &- (k_3\boldsymbol{A}^{-1}\boldsymbol{k}_v + k\boldsymbol{A}^{-1})\mathrm{sig}^{\alpha_1+\beta-1}(\hat{\boldsymbol{e}}_3) + \boldsymbol{\Delta}_s \end{aligned} \quad (48)$$

where $\boldsymbol{\Delta}_s = -\boldsymbol{J}_{Eg}\tilde{\boldsymbol{x}}_3 + \dot{\boldsymbol{J}}_{Eg}\dot{\boldsymbol{\theta}} - \dot{\hat{\boldsymbol{J}}}_{Eg}\hat{\boldsymbol{x}}_2$ is caused by the observer errors. $\boldsymbol{\Delta}_s$ is omitted and the acceleration $\ddot{\boldsymbol{x}}_\tau'$ is approximated by $\ddot{\boldsymbol{x}}_s' = \ddot{\boldsymbol{x}}_\tau' - \boldsymbol{\Delta}_s$. On the other hand, according to (4), we can obtain $\dot{\boldsymbol{x}}_e' = \boldsymbol{J}_E\left[\boldsymbol{J}_b\dot{\boldsymbol{x}}_0 + \boldsymbol{J}_m\dot{\boldsymbol{\theta}}\right]$ and

$$\ddot{\boldsymbol{x}}_e' = \boldsymbol{J}_E\left[\boldsymbol{J}_b\ddot{\boldsymbol{x}}_0 + \boldsymbol{J}_m\ddot{\boldsymbol{\theta}}\right] + \boldsymbol{V}_s \quad (49)$$

where $\boldsymbol{V}_s = \left[\dot{\boldsymbol{J}}_E\boldsymbol{J}_b + \boldsymbol{J}_E\dot{\boldsymbol{J}}_b\right]\dot{\boldsymbol{x}}_0 + \left[\dot{\boldsymbol{J}}_E\boldsymbol{J}_m + \boldsymbol{J}_E\dot{\boldsymbol{J}}_m\right]\dot{\boldsymbol{\theta}}$. Combining (48) and (49), $\ddot{\boldsymbol{\theta}}_\tau$ can be obtained by solving the following equation:

$$\ddot{\boldsymbol{x}}_m = \boldsymbol{J}_m\ddot{\boldsymbol{\theta}}_\tau \quad (50)$$

where $\ddot{\boldsymbol{x}}_m = \boldsymbol{J}_E^{-1}\left(\ddot{\boldsymbol{x}}_s' - \boldsymbol{V}_s\right) - \boldsymbol{J}_b\ddot{\boldsymbol{x}}_0$.

Almost all the methods of solving $\dot{\boldsymbol{x}}_e = \boldsymbol{J}_m\dot{\boldsymbol{\theta}}$ can be employed to deal with (50). In this paper, the body-fixed frames of the space robot is defined as Fig. 2 (when all of the joint angles are zero). Since the last three rotation axes intersect at one point (the wrist is spherical), the manipulator can be called the PUMA-type manipulator. The SSDR method proposed in [32] is practical to handle singularities for a PUMA-type manipulator. Apply SSDR to the space robot shown in Fig. 2 and the solution to (50) can be obtained by the following operations.

The wrist value corresponding to $\ddot{\boldsymbol{x}}_m$ is

$$\ddot{\boldsymbol{x}}_w^0 = \boldsymbol{R}_0^{\mathrm{T}}\boldsymbol{U}^{-1}\ddot{\boldsymbol{x}}_m \quad (51)$$

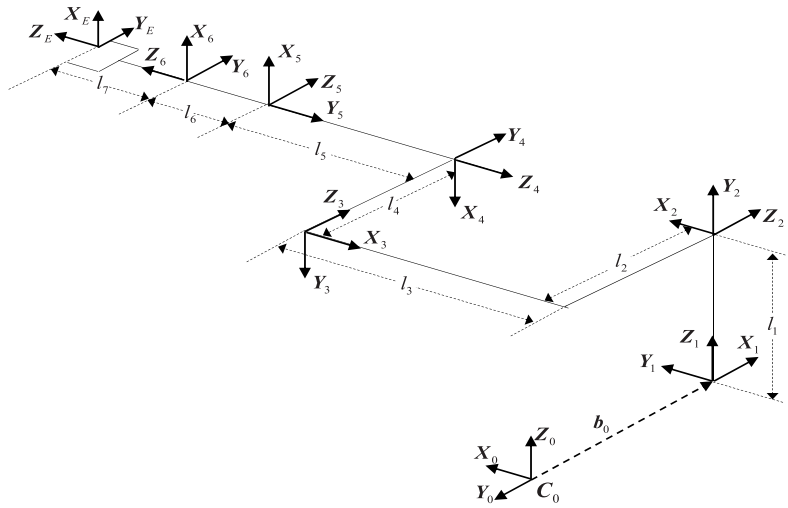


Fig. 2. The body-fixed frames of the space robotic system.

where $\boldsymbol{R}_0 \in \boldsymbol{R}^{3\times3}$ is the rotation matrix of the base frame relative to the inertial frame, and

$$\boldsymbol{U} = \begin{bmatrix} \boldsymbol{E}_3 & -d_6\boldsymbol{Z}_6^{\times} \\ \boldsymbol{0} & \boldsymbol{E}_3 \end{bmatrix}$$

Denote d_6 as $d_6 = \|\boldsymbol{a}_5+\boldsymbol{b}_5\|+\|\boldsymbol{a}_6+\boldsymbol{b}_6\|$. $\boldsymbol{Z}_i$ is the unit vector representing the rotation direction of joint i. Define $\boldsymbol{a}_p^0 = \begin{bmatrix}\ddot{x}_{w1}^0 & \ddot{x}_{w2}^0 & \ddot{x}_{w3}^0\end{bmatrix}^{\mathrm{T}}$, $\boldsymbol{a}_\omega^0 = \begin{bmatrix}\ddot{x}_{w4}^0 & \ddot{x}_{w5}^0 & \ddot{x}_{w6}^0\end{bmatrix}^{\mathrm{T}}$, $\ddot{\boldsymbol{\theta}}_f = \begin{bmatrix}\ddot{\theta}_{\tau1} & \ddot{\theta}_{\tau2} & \ddot{\theta}_{\tau3}\end{bmatrix}^{\mathrm{T}}$, and $\ddot{\boldsymbol{\theta}}_l = \begin{bmatrix}\ddot{\theta}_{\tau4} & \ddot{\theta}_{\tau5} & \ddot{\theta}_{\tau6}\end{bmatrix}^{\mathrm{T}}$. The wrist value also can be calculated as

$$\begin{bmatrix} \boldsymbol{a}_p^0 \\ \boldsymbol{a}_\omega^0 \end{bmatrix} = \begin{bmatrix} \boldsymbol{J}_{11} & \boldsymbol{0} \\ \boldsymbol{J}_{21} & \boldsymbol{J}_{22} \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{\theta}}_f \\ \ddot{\boldsymbol{\theta}}_l \end{bmatrix} \tag{52}$$

where $\boldsymbol{J}_{11}$, $\boldsymbol{J}_{21}$, and $\boldsymbol{J}_{22}$ can be obtained in [32]. The approximate angular acceleration $\ddot{\boldsymbol{\theta}}_\tau$ can be derived by

$$\begin{aligned} \ddot{\boldsymbol{\theta}}_f &= \boldsymbol{J}_{11}^{-1}\boldsymbol{a}_p^0 \\ \ddot{\boldsymbol{\theta}}_l &= \boldsymbol{J}_{22}^{-1}\left(\boldsymbol{a}_\omega^0 - \boldsymbol{J}_{21}\ddot{\boldsymbol{\theta}}_f\right) \end{aligned} \tag{53}$$

Therefore, the singularities occur, when $\boldsymbol{J}_{11}$ or $\boldsymbol{J}_{22}$ is not invertible. Similarly to [32], the inverse of $\boldsymbol{J}_{11}$ or $\boldsymbol{J}_{22}$ can be expressed as

$$\begin{aligned} \boldsymbol{J}_{11}^{-1} &= \mathrm{diag}\left(\frac{1}{k_s}, \frac{1}{k_b}, \frac{1}{k_b}\right)\boldsymbol{F}_1 \\ \boldsymbol{J}_{22}^{-1} &= \frac{1}{k_w}\boldsymbol{F}_2 \end{aligned} \tag{54}$$

where $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$ are two ordinary matrices without infinite elements, and

$$\begin{aligned} k_s &= -d_4 s_{23} + l_3 c_2 \\ k_b &= d_4 s_3 \\ k_w &= -s_5 \end{aligned}$$

Define d_4 and l_3 as $d_4 = \|\boldsymbol{a}_4 + \boldsymbol{b}_4\|$ and $l_3 = |a_{2x} + b_{2x}|$. c_i and s_i denote $\cos\theta_i'$ and $\sin\theta_i'$, respectively. c_{ij} and s_{ij} are computed by $c_{ij} = \cos\left(\theta_i' + \theta_j'\right)$ and $s_{ij} = \sin\left(\theta_i' + \theta_j'\right)$. $\boldsymbol{\theta}' = \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$ is satisfied, where $\Delta\boldsymbol{\theta} = \begin{bmatrix}\pi & -\pi & -\frac{\pi}{2} & 0 & \pi & 0\end{bmatrix}^{\mathrm{T}}$.

According to (54), $k_s = 0$ or $k_b = 0$ implies that $\boldsymbol{J}_{11}$ is not invertible. $k_w = 0$ causes that $\boldsymbol{J}_{22}$ is not invertible. Therefore, parameters leading to singularities have been found and separated successfully. A method based on damped reciprocal is employed to deal with $1/k_s$, $1/k_b$, and $1/k_w$. Finally, each element of $\ddot{\boldsymbol{\theta}}_\tau$ can be computed as

$$\begin{aligned} \ddot{\theta}_{\tau1} &= \frac{k_s}{k_s^2+\lambda_s^2}\tilde{a}_{p2} \\ \ddot{\theta}_{\tau2} &= -\frac{k_b d_4}{l_3(k_b^2+\lambda_b^2)}\tilde{a}_{p3}^3 \\ \ddot{\theta}_{\tau3} &= -\frac{k_b}{k_b^2+\lambda_b^2}\left(\frac{k_q}{l_3}\tilde{a}_{p1} - \frac{k_t}{l_3}\tilde{a}_{p3}\right) \\ \ddot{\theta}_{\tau4} &= \frac{k_w}{k_w^2+\lambda_w^2}\Delta a_{w1} \\ \ddot{\theta}_{\tau5} &= \Delta a_{w2} \\ \ddot{\theta}_{\tau6} &= -\frac{k_w c_5}{k_w^2+\lambda_w^2}\Delta a_{w1} + \Delta a_{w3} \end{aligned} \tag{55}$$

where

$$\begin{aligned} \tilde{\boldsymbol{a}}_p &= \boldsymbol{Q}_1\boldsymbol{a}_p^0 \\ \Delta\boldsymbol{a}_\omega &= \boldsymbol{Q}_2\left(\boldsymbol{a}_\omega^0 - \boldsymbol{J}_{21}\ddot{\boldsymbol{\theta}}_f\right) \\ k_q &= -l_3 c_3 \\ k_t &= -d_4 + l_3 s_3 \end{aligned}$$

$\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ can be obtained in [32]. λ_s, λ_b and λ_w are termed damped coefficients. They are given by

$$\lambda_x^2 = \begin{cases} \lambda_0\left(1 - \frac{|k_x|}{\varepsilon_x}\right)^2 & |k_x| \le \varepsilon_x \\ 0 & |k_x| > \varepsilon_x \end{cases} \qquad x = s, b, w \tag{56}$$

where $\varepsilon_x (x = s, b, w)$ are the thresholds of the corresponding singularity parameters.

According to the dynamic equation (3), if the dynamic singularities occur, the large control input τ in (40) can be replaced by

$$\boldsymbol{\tau}_s = \boldsymbol{H}_g\ddot{\boldsymbol{\theta}}_\tau - \boldsymbol{H}_g\hat{\boldsymbol{x}}_3 \tag{57}$$

In summary, when the dynamic singularities are met, the approximate angular acceleration $\ddot{\boldsymbol{\theta}}_\tau$ can be obtained by (55), and then the control input $\boldsymbol{\tau}_s$ in (57) is adopted instead of the original input.

Remark 4. Multiplying both sides of (50) by a small time δt, the equation is transformed into $\delta\dot{\boldsymbol{x}}_m = \boldsymbol{J}_m\delta\dot{\boldsymbol{\theta}}_\tau$. It can represent that, under the assumption of the same configuration of the space robots, if there exist the joint-velocity errors $\delta\dot{\boldsymbol{\theta}}_\tau$, the velocity errors of the end-effector are $\delta\dot{\boldsymbol{x}}_m$.

Remark 5. With SSDR, the solution to (50) can be calculated by (55) directly. Compared with the traditional damped least-squares (DLS) method, since the estimation of the minimum singularity of the Jacobian matrix is not required, the computation load of SSDR

Table 2
Parameters of the observer, the controller and the simulations.

Parameter	Value
Fixed-time extended state observer	$p = 0.9, q = 1.2, k_{o1} = 12, k_{o2} = 35$ $k_{o3} = 5, T_s = 0.1$ s
Fixed-time controller	$\alpha = 0.8,\ \beta = 1.2,\ k_a = 0.2,\ k = 6$ $k_1 = 1, k_2 = k_3 = 2$
Parameters of handling singularities	$\varepsilon_3 = 0.01, \varepsilon_s = \varepsilon_b = 0.05, \varepsilon_w = 0.15$ $\lambda_0 = 0.08$
External disturbance	$\tau_{di} = 1 \times 10^{-4} \sin\left(\frac{\pi}{5}t\right), i = 1, 2, \ldots, 6$

is reduced. However, this method is only suitable for a PUMA-type manipulator, which consists of a 3-DOF forearm with a 3-DOF spherical wrist. Moreover, in terms of space robots with different structures, the representations of the solutions are different.

4. Simulation results

The space robotic system in this section is composed of a spacecraft and a 6 DOF manipulator as shown in Fig. 1 and Fig. 2. The mass and inertia properties of the space robot are given in [39]. The parameters of the observer-based fixed time controller and simulations are listed in Table 2. The initial position and attitude of the desired trajectories are $\boldsymbol{p}_d(0) = [1.7559\ 0.0196\ 0.9276]^{\mathrm{T}}$ (m) and $\boldsymbol{q}_d(0) = [0.7469\ \ -0.0589\ 0.7293]^{\mathrm{T}}$. The end-effector is required to track the trajectories with following constant velocities.

$$\dot{\boldsymbol{p}}_d(t) = -[0.0067 \quad 0.0033 \quad 0.0133]^{\mathrm{T}}\ (\mathrm{m/s})$$
$$\boldsymbol{\omega}_d(t) = [-0.0138 \quad 0.0325 \quad 0.0041]^{\mathrm{T}}\ (\mathrm{rad/s})$$

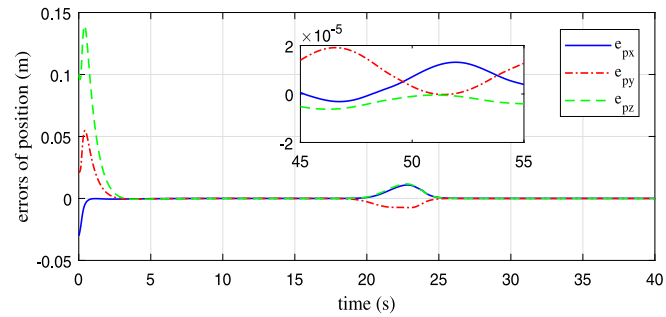
The initial joint angles and velocities of the space robot are $\boldsymbol{\theta}(0) = [3.35 \quad 53.82 \quad -84.67 \quad -6.96 \quad 40.84 \quad 1.65]^{\mathrm{T}}$ (°) and $\dot{\boldsymbol{\theta}}(0) = \mathbf{0}$. The corresponding states of the end-effector are $\boldsymbol{p}_e(0) = [1.7260 \quad 0.0412 \quad 1.0276]^{\mathrm{T}}$ (m) and $\boldsymbol{q}_e(0) = [0.6714 \quad -0.0527 \quad 0.7977]^{\mathrm{T}}$.

The simulation results are shown in Figs. 3–9. The control errors including position and attitude of the end-effector are shown in Fig. 3. From Fig. 3, the system with the proposed controller is stable at 12 s, firstly. At 18 s, the system breaks the original stable state due to dynamic singularities (see Remark 7 for an explanation), and reaches stable at 30 s, again. The control accuracies of the position and the attitude are 2×10^{-5} m and 2×10^{-4}, respectively. The performance of the fixed-time observer is shown in Figs. 4 and 5. The observer accuracies of the joint angles and the joint velocities are 1×10^{-3}° and 0.05°/s, respectively. To illustrate the movement of the end-effector further, the actual and desired trajectories are compared in Figs. 7 and 8. Therefore, the performance of the controller and the observer is satisfactory.

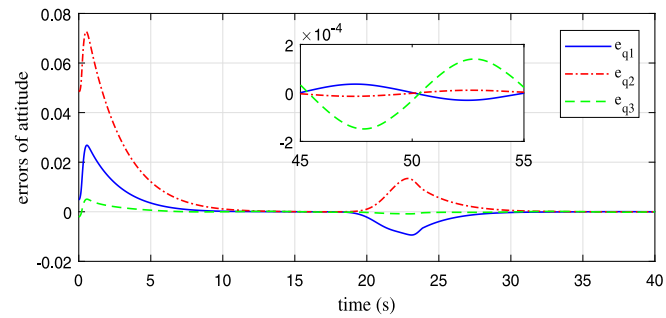
Remark 6. From Figs. 3, 7 and 8, at the initial stage, the response of the controller is not very good. The reason is that the required estimation precision of the joint velocity has not been achieved until 1.5 s as shown in Figs. 4 and 5.

Remark 7. As shown in Fig. 9, the absolute value of $|\boldsymbol{J}_g|$ is less than the threshold ε_3 from 18.0 s to 23.8 s, which means that there are dynamic singularities during the time period. The original input (40) is replaced by the control input (57) when the singularities occur. It leads to the deviation as shown in Figs. 3 and 7. After 23.8 s, the system is no longer affected by the singularities, and with the proposed controller (57), the tracking errors begin to decrease. This is also why the control input becomes larger at around 23.8 s in Fig. 6. Finally, the system reaches stable at 30 s again as shown in Fig. 3.

The proposed controller is compared with the controller not considering dynamic singularities, which means that the controller (40) is still employed even if the singularities occur. From Fig. 10, taking $\dot{\theta}_4$ as an example, the maximum is 45°/s with the method of this paper, while



(a) Errors of position $\boldsymbol{e}_p$



(b) Errors of attitude $\boldsymbol{e}_q$

Fig. 3. Control errors of position and attitude of the end-effector.

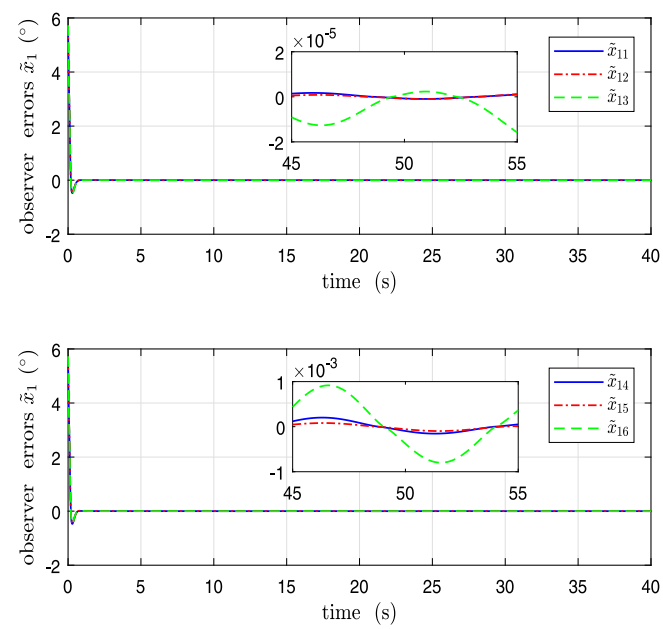


Fig. 4. Observer errors of joint angles.

the maximum is 110°/s with the controller not considering dynamic singularities. Therefore, based on this comparison and Remark 7, we can conclude that the SSDR method is effective to deal with dynamic singularities.

To further demonstrate the effectiveness of the FXPAC, comparisons are performed with a finite-time control method proposed in [40],

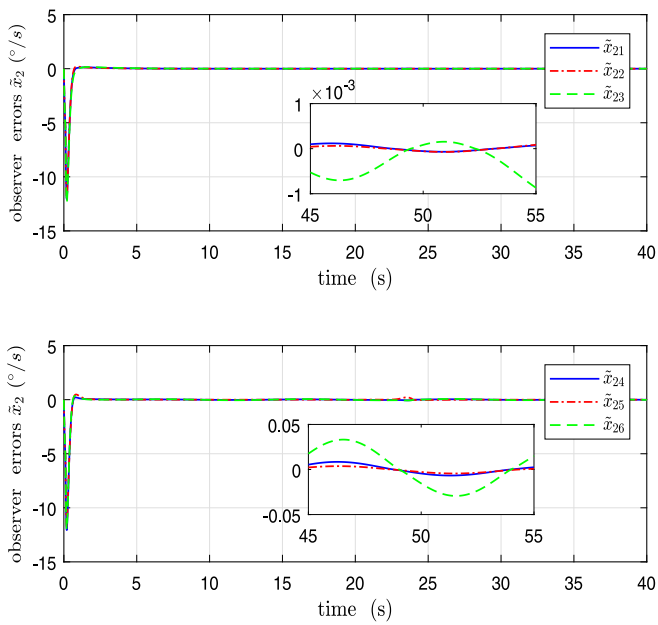


Fig. 5. Observer errors of joint velocities.

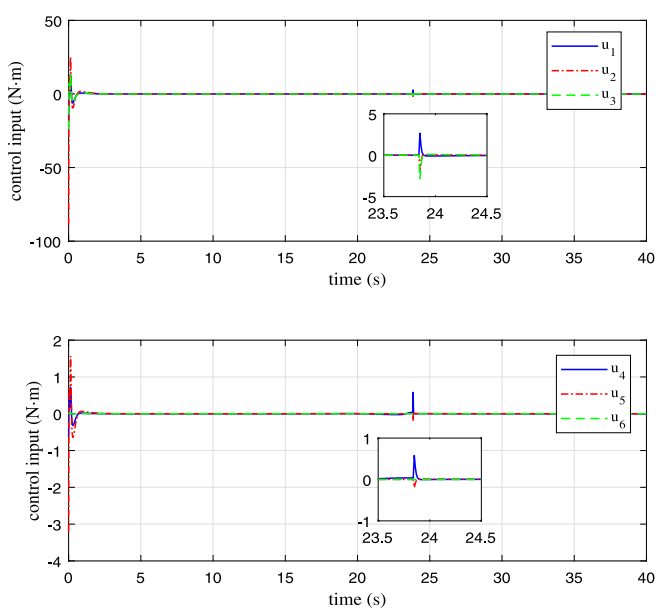


Fig. 6. Control inputs.

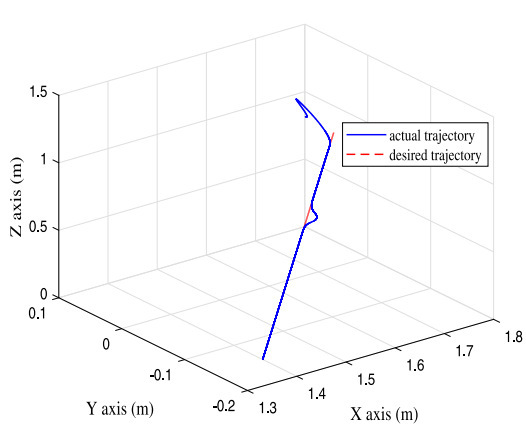


Fig. 7. Desired and actual position of the end-effector.

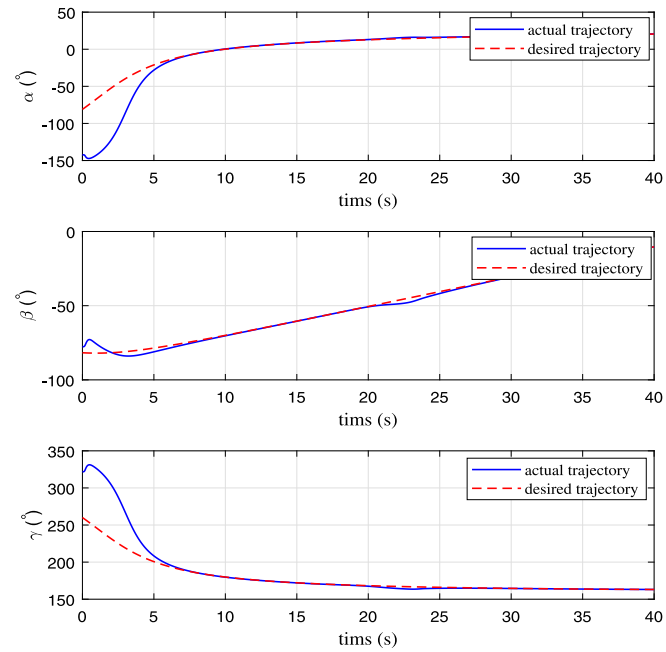


Fig. 8. Desired and actual attitude represented in Euler angles (Z–Y–X) of the end-effector.

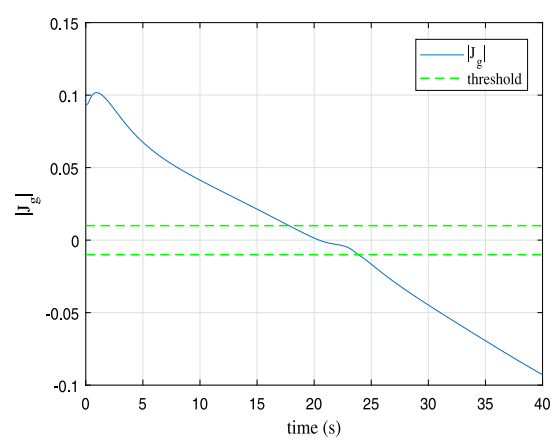


Fig. 9. Determinant of matrix $\boldsymbol{J}_g$.

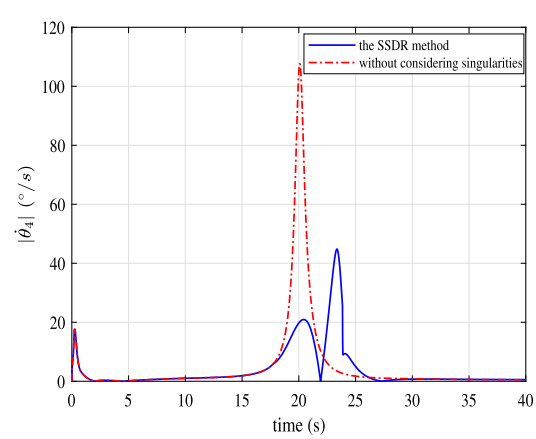


Fig. 10. Trajectory of $\dot{\theta}_4$ using controller not considering dynamic singularities.

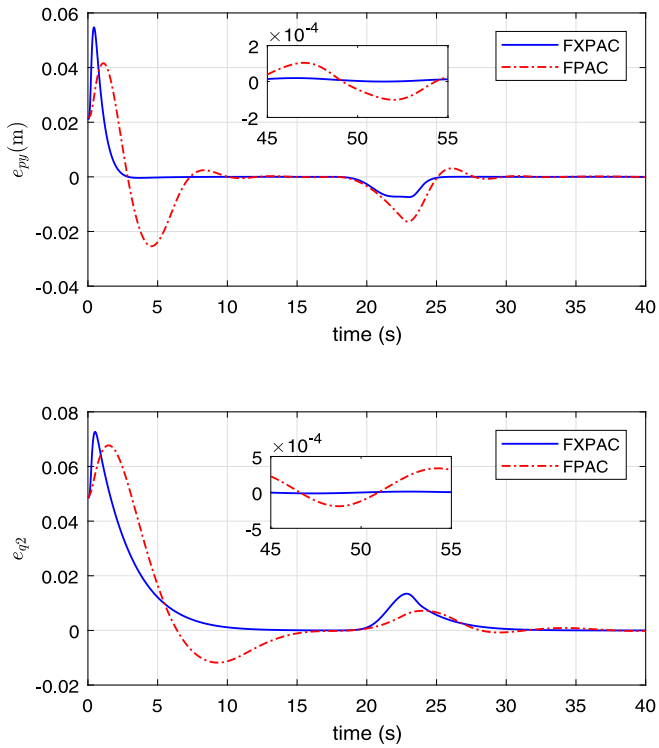


Fig. 11. Errors using FXPAC and FPAC.

which can be expressed as

$$\boldsymbol{\tau} = -\boldsymbol{H}_g\hat{\boldsymbol{x}}_3 + \boldsymbol{H}_g\boldsymbol{J}_{Eg}^{-1}\left[\ddot{\boldsymbol{x}}_d' - \dot{\hat{\boldsymbol{J}}}_{Eg}\hat{\boldsymbol{x}}_2 - \iota_1\boldsymbol{A}^{-1}\boldsymbol{k}_v\left(\boldsymbol{e}_2^{\hbar} + \iota_2^{\hbar}\boldsymbol{e}_1\right)^{\frac{2}{\hbar}-1}\right]$$

where $\hbar = \frac{\hbar_1}{\hbar_2} = \frac{21}{19}$, $\iota_1 = 0.5$, and $\iota_2 = 0.8$. The above control is termed the finite-time position and attitude control (FPAC) here. Taking e_{py} and e_{q2} as examples, the comparison results of FXPAC and FPAC are shown in Fig. 11. Compared with FPAC, FXPAC can achieve better control accuracies, a shorter convergence time, and fewer oscillations. It can be seen that the proposed FXPAC can provide better control performance than FPAC. Based on the above, the simulation results can illustrate the effectiveness of the observer-based fixed-time control considering dynamic singularities.

5. Conclusions

This paper studied a novel FXPAC method based on FXESO for a free-floating space robot. Both the joint velocity and the lumped disturbance were estimated by the FXESO. With the FXPAC method, the tracking errors of the position and the attitude of the end-effector could converge to a neighborhood of the origin within fixed time, independent of the initial conditions. The MRPs were adopted to represent the attitude during the control design. Moreover, the SSDR method was employed to avoid large control inputs or large joint velocities, when the dynamic singularities occurred. Simulation results have demonstrated the effectiveness of the proposed control method.

Further studies need to be conducted for space robots. The control performance is impacted by actuator saturation [9], actuator faults [3], and the dead zone [41]. Therefore, they will also become a part of our future studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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